

CALCULUS I | AP CALCULUS AB/BC

Optimization

A Structured Problem-Solving Framework

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Level: Grades 11–12 / College Intro

| 60–75 Minutes

Prereqs: Derivative Tests, Critical Points, Extreme Value Theorem

Artifact: Instructional Sample

Central Idea: Transforming verbal optimization scenarios into single-variable calculus models by isolating objectives, leveraging constraints, and verifying global extrema.

Instructional Focus & Objectives

Optimization problems represent a quintessential application of differential calculus: determining the “best” configuration—maximum area, minimal material, least cost, or shortest travel time—subject to physical and financial boundaries. While students often feel overwhelmed by the variety of geometric shapes and real-world contexts, every optimization scenario hinges on a shared structural core: distinguishing what we seek to optimize from the limitations that constrain us.

- **Conceptual Understanding:** Clearly differentiate between the **Objective Function** (the quantity to be maximized or minimized) and the **Constraint Equation** (the physical restriction binding the variables together).
- **Mathematical Modeling:** Translate verbal descriptions into labeled diagrams, define explicit algebraic variables, establish feasible physical domains, and eliminate auxiliary variables via algebraic substitution.
- **Analytical Rigor:** Utilize first and second derivatives to locate critical points and systematically apply the Extreme Value Theorem (Closed Interval Method) or derivative tests to rigorously prove that a candidate point is indeed an absolute extremum.
- **Contextual Interpretation:** Clearly distinguish between the *optimizing input value* (e.g., cut depth x) and the *optimal value achieved* (e.g., maximum volume $V_{\max}$), stating final answers with complete physical units.

Anticipated Misconceptions & Pedagogical Interventions

Critical Error 1: Conflating the Objective Function with the Constraint

Misconception: Attempting to take the derivative of the constraint equation (e.g., differentiating the perimeter $2l + 2w = 60$ to get $2 + 2\frac{dw}{dl} = 0$) rather than differentiating the objective quantity (area $A = lw$).

Instructional Strategy: Have students explicitly write two labeled headers at the top of their page before doing any algebra:

OBJECTIVE (To Optimize): $A = l \times w$

CONSTRAINT (Fixed Limit): $2l + 2w = 60 \implies l = 30 - w$

Reinforce that we only differentiate the objective function; the constraint exists solely to reduce the objective to a single independent variable.

Critical Error 2: Ignoring the Feasible Physical Domain

Misconception: Treating the problem as an unconstrained algebraic exercise on $(-\infty, \infty)$ and failing to determine the domain boundaries dictated by geometry (e.g., side lengths must be positive, or corner cut size x cannot exceed half the container width).

Mathematical Reality: Without domain bounds, students cannot apply the Closed Interval Method, may accept physically impossible negative critical numbers, or miss potential boundary extrema.

Pedagogical Prompt: “What is the smallest possible value this variable could physically take without destroying the shape? What is the absolute largest?”

Critical Error 3: Assuming Any Critical Point is Automatically the Answer

Misconception: Finding a critical point where $f'(x) = 0$ and immediately writing it down as the answer without verifying whether it represents a maximum, a minimum, or an inflection point, and without testing endpoints.

Instructional Strategy: Require explicit candidate justification: either an endpoint/critical-point evaluation table (for closed intervals) or a First/Second Derivative Test sign chart (for open intervals).

Lesson Launch: The Geometry of Trade-Offs

Teacher Guidance & Instructional Launch (7–10 Minutes)

The Concrete Scenario: Present students with a loop of string of fixed length 60 cm. Ask two volunteers to pull it into different rectangular shapes:

- A long, narrow rectangle: $29 \text{ cm} \times 1 \text{ cm} \implies \text{Area} = 29 \text{ cm}^2$.
- A more balanced rectangle: $20 \text{ cm} \times 10 \text{ cm} \implies \text{Area} = 200 \text{ cm}^2$.
- A square: $15 \text{ cm} \times 15 \text{ cm} \implies \text{Area} = 225 \text{ cm}^2$.

The Driving Question: “Every rectangle has the exact same perimeter (60 cm). Why does the enclosed area vary so drastically? Can we prove which shape encloses the absolute maximum area?”

The Key Insight: As width increases, length must decrease to satisfy the perimeter budget. Area is the product of these competing dimensions. Optimization is the mathematical art of balancing trade-offs.

A Structured Problem-Solving Framework

Mathematical modeling is not mechanical formula-plugging; mathematical judgment is essential at every stage. However, having a structured framework provides students with a dependable starting point when facing intimidating word problems.

The Optimization Problem-Solving Framework

Step 1: Identify the Goal & Assign Variables: Read the context carefully. Identify exactly what quantity is to be maximized or minimized. Assign clear variable letters and physical units to all relevant dimensions.

Step 2: Sketch & Label the Geometry: Draw a representative diagram illustrating the physical system. Mark known constants and label dynamic variables.

Step 3: Formulate the Objective Function: Write an equation representing the quantity to be optimized in terms of the problem variables (e.g., $A = lw$, $V = xyz$, $C = 10A_{\text{base}} + 6A_{\text{sides}}$).

Step 4: Identify Constraint Equation(s) & Eliminate Variables: Identify any given limitations (fixed perimeter, total volume, material budget). Use these algebraic constraints to express auxiliary variables in terms of a single primary variable, substituting back into the objective function to produce a single-variable model $f(x)$.

Step 5: Establish the Feasible Domain: Determine the physical domain $[a, b]$ or (a, b) of the independent variable based on physical constraints (e.g., non-negative lengths, geometric fit).

Step 6: Find Critical Numbers via Differentiation: Compute the derivative $f'(x)$. Locate all critical points in the domain where $f'(x) = 0$ or where $f'(x)$ fails to exist.

Step 7: Test Candidates & Prove Global Extrema:

- *Closed Interval* $[a, b]$: Evaluate $f(x)$ at all interior critical points and both endpoints (Closed Interval Method / Extreme Value Theorem).
- *Open Interval* (a, b) : Use the First Derivative Test (sign analysis of f') or Second Derivative Test ($f''(c) < 0 \implies$ local maximum, $f''(c) > 0 \implies$ local minimum) to establish concavity and global behavior.

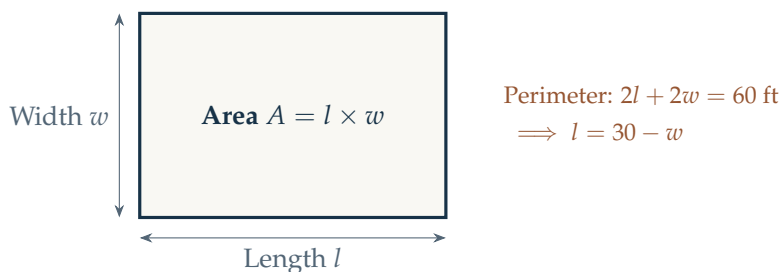
Step 8: Interpret the Result & State the Final Conclusion: Carefully re-read the original prompt. Does it ask for the dimensions, the maximum value, or the minimal cost? State the answer in a complete sentence with correct compound units.

Guided Modeling & Worked Examples

Example 1: Maximizing Rectangular Area with Fixed Perimeter

Problem Statement

A gardener has 60 feet of fencing to enclose a rectangular vegetable plot. Find the dimensions of the rectangle that maximize the enclosed area, and determine the maximum area.

**Step-by-Step Solution:**

Step 1: Identify Goal: Maximize rectangular area A . Let length = l (ft), width = w (ft).

Step 2: Diagram: A rectangle with side lengths l and w .

Step 3: Objective Function: $A = l \cdot w$.

Step 4: Constraint Equation: Total fencing is 60 ft:

$$2l + 2w = 60 \implies l + w = 30 \implies l = 30 - w$$

Substitute into the objective function:

$$A(w) = (30 - w)w = 30w - w^2$$

Step 5: Feasible Domain: Length and width must be non-negative: $w \geq 0$ and $l = 30 - w \geq 0 \implies w \leq 30$. The feasible domain is the closed interval $[0, 30]$.

Step 6: Critical Points:

$$A'(w) = 30 - 2w = 0 \implies 2w = 30 \implies w = 15 \text{ ft}$$

The derivative exists everywhere; $w = 15$ is the sole critical point in $(0, 30)$.

Step 7: Candidate Testing (Closed Interval Method):

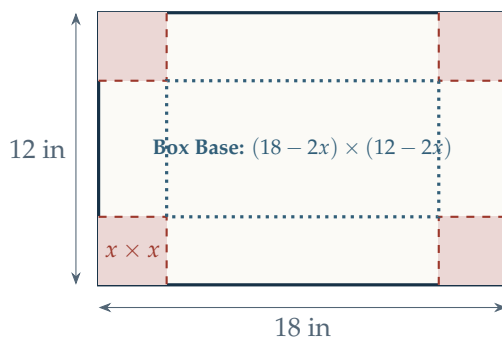
Candidate w	Area $A(w) = 30w - w^2$	Classification
$w = 0$ (endpoint)	$A(0) = 0 \text{ ft}^2$	Absolute Minimum
$w = 15$ (critical point)	$A(15) = 30(15) - (15)^2 = 225 \text{ ft}^2$	Absolute Maximum
$w = 30$ (endpoint)	$A(30) = 0 \text{ ft}^2$	Absolute Minimum

Step 8: Final Interpretation: The enclosed area is maximized when width $w = 15$ ft and length $l = 30 - 15 = 15$ ft (a square configuration). The maximum achievable area is 225 ft^2 .

Example 2: Fabricating an Open-Top Box (Geometry with Cuts)**Problem Statement**

An open-top storage box is to be constructed from a rectangular sheet of cardboard measuring 12 inches by 18 inches by cutting congruent squares of side length x from each of the four corners and folding up the resulting flaps. What cut size x maximizes the interior volume of the resulting

box?



Step-by-Step Solution:

Step 1: Identify Goal: Maximize the box volume V . Variable: cut size x (in).

Step 2: Sketch Geometry: After folding, the 3D dimensions of the open box are:

$$\text{Height } h = x, \quad \text{Length } L = 18 - 2x, \quad \text{Width } W = 12 - 2x$$

Step 3: Objective Function:

$$V(x) = x(18 - 2x)(12 - 2x) = x(216 - 60x + 4x^2) = 4x^3 - 60x^2 + 216x$$

Step 4: Constraints: The relationship is already reduced to the single variable x through the sheet's fixed dimensions.

Step 5: Feasible Domain: Each dimension must be strictly non-negative:

$$x \geq 0, \quad 12 - 2x \geq 0 \implies x \leq 6, \quad 18 - 2x \geq 0 \implies x \leq 9$$

The more restrictive condition dictates the feasible interval: $x \in [0, 6]$.

Step 6: Differentiate to Find Critical Points:

$$V'(x) = 12x^2 - 120x + 216 = 12(x^2 - 10x + 18)$$

Setting $V'(x) = 0$ yields the quadratic equation $x^2 - 10x + 18 = 0$. By the quadratic formula:

$$x = \frac{10 \pm \sqrt{(-10)^2 - 4(1)(18)}}{2} = \frac{10 \pm \sqrt{100 - 72}}{2} = \frac{10 \pm \sqrt{28}}{2} = 5 \pm \sqrt{7}$$

Evaluating the two roots:

- $x_1 = 5 + \sqrt{7} \approx 5 + 2.646 = 7.646$ in (Extraneous: lies outside $[0, 6]$).
- $x_2 = 5 - \sqrt{7} \approx 5 - 2.646 = 2.354$ in (Feasible: lies within $(0, 6)$).

Step 7: Candidate Testing (Closed Interval Method):

Candidate x	Volume $V(x)$	Physical Reasoning
$x = 0$ (no cut)	$V(0) = 0 \text{ in}^3$	Flat cardboard sheet; zero height
$x = 5 - \sqrt{7} \approx 2.354$	$V(2.354) \approx 228.3 \text{ in}^3$	Absolute Maximum Volume
$x = 6$ (maximum cut)	$V(6) = 0 \text{ in}^3$	Cardboard cut entirely through; zero width

Step 8: Final Interpretation: To maximize the volume of the box, corner squares with side length $x = 5 - \sqrt{7} \approx 2.35$ inches should be removed. The resulting box has dimensions approximately $2.35 \text{ in} \times 7.29 \text{ in} \times 13.29 \text{ in}$, yielding a maximum volume of approximately 228.3 cubic inches.

Teacher Commentary: Emphasizing Mathematical Judgment over Routine

Notice how Step 5 (Feasible Domain) eliminated $x = 5 + \sqrt{7} \approx 7.65$ in without cumbersome arithmetic. If a student ignores the domain and tests $x = 7.65$, they compute a “negative volume” because $12 - 2(7.65) = -3.3$ in. Emphasize to students: *The algebra does not know you are cutting cardboard; only your domain restriction enforces reality!*

Class Practice & Scaffolded Exercises

- Number Product:** Find two positive real numbers whose sum is 24 and whose product is as large as possible.
- Riverfront Enclosure (Single Constraint):** A livestock farmer wishes to fence a rectangular pasture along a straight riverbank. No fencing is needed along the water. With 2400 feet of fencing available, find the dimensions that maximize the grazing area.
- Industrial Packaging Design (Cost Minimization):** A rectangular storage container with an open top must hold a volume of 20 m^3 . The base length is twice the base width. Heavy-duty material for the base costs $\$10/\text{m}^2$, while side material costs $\$6/\text{m}^2$. Find the dimensions that minimize the total construction cost.
- Beverage Can Design (Surface Area):** A cylindrical aluminum can must hold 500 mL (500 cm^3) of liquid. Determine the ratio of height h to radius r that minimizes total surface area (top and bottom discs included).

Formative Assessment: Individual Exit Task

Name: _____ Date: _____

Task A (Applied Modeling): A rectangle is inscribed inside a semicircle of radius $R = 5$, with its base resting along the diameter.

- Draw and label a coordinate diagram, letting the upper-right vertex be (x, y) on the circle $x^2 + y^2 = 25$.
- Formulate the objective function for area A in terms of x alone, stating its physical domain.
- Find the critical point and prove whether it yields an absolute maximum or minimum.
- State the optimal dimensions and the maximum area.

Task B (Conceptual Reasoning): In optimization, why is it essential to reduce the objective function to a *single* independent variable before differentiating? What would happen if you differentiated $A = l \cdot w$ directly with respect to l without substituting the constraint?

Differentiation & Pedagogical Supports

Targeted Support (Struggling Learners): • **Two-Column Model Sheet:** Provide an organizer separating the left column (“What do I want to optimize?”) from the right column (“What is holding me back?”).

- **Numerical Exploration Table:** Before writing algebraic formulas, have students test 3 explicit numbers (e.g., width = 5, 10, 15) to see the quadratic curve take shape concretely.
- **Unit Tracking:** Require students to write units beside critical numbers to ensure they do not confuse an input dimension ($x = 2.35$ in) with the optimal output ($V = 228.3$ in³).

Enrichment & Extension (Advanced Learners): • **Generalized Can Geometry:** Prove that for *any* volume V , the optimal cylinder minimizing surface area always has height equal to diameter ($h = 2r$).

- **Economic Margin Analysis:** Connect the critical point $C'(x) = 0$ to marginal cost and revenue in introductory microeconomics.

Curricular Context & References

This instructional sample aligns with standard secondary and introductory collegiate calculus curricula (College Board AP Calculus AB/BC Topic 5.10 and 5.11: Solving Optimization Problems).

- Stewart, J. *Calculus: Early Transcendentals*, 8th ed.; Cengage Learning, 2016; Chapter 4.7 (Optimization Problems).
- Larson, R.; Edwards, B. H. *Calculus*, 11th ed.; Cengage Learning, 2018; Section 3.7.
- College Board. *AP Calculus AB and BC Course and Exam Description*, Effective Fall 2020; Unit 5: Analytical Applications of Differentiation.